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**Gilford et al.**

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(54) **STRAWBERRY PLANT NAMED ‘DRISCOLL DESTIN’**

(50) Latin Name: *Fragaria*×*ananassa*  
Varietal Denomination: **Driscoll Destin**

(76) Inventors: **Kristie L. Gilford**, 1525 N. Dover Rd.,  
Dover, FL (US) 33527; **Bruce D.**  
**Mowrey**, 266 Webb Rd., Watsonville,  
CA (US) 95076; **JoAnne Coss**, 217  
Katherine Ave., Salinas, CA (US) 93901

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*Primary Examiner*—Anne Marie Grunberg  
*Assistant Examiner*—Annette H Para  
(74) *Attorney, Agent, or Firm*—Jones Day

(57) **ABSTRACT**

This invention relates to a new and distinct variety of  
strawberry named ‘Driscoll Destin.’ The variety is similar to  
the varieties ‘Biscayne’ and ‘Key Largo.’ The variety is  
distinguished from ‘Biscayne’ and ‘Key Largo,’ in  
particular, by its shorter plant height, larger flower diameter,  
an orange-red external fruit color, greater yield, weak leaf  
glossiness, very dense stipule pubescence, and a small  
hollow fruit center.

**5 Drawing Sheets**

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1. Latin name of the genus and species of the plant  
claimed: The variety is botanically identified as *Fragaria*×  
*ananassa*.

1.1 Variety denomination: The strawberry variety  
denomination is ‘Driscoll Destin’.

**2. BACKGROUND OF THE INVENTION**

The new variety originated as a result of a controlled cross  
between the strawberry plants ‘73D144’ (unpatented  
variety) and ‘88E94’ (unpatented variety) in an ongoing  
breeding program, and was discovered as a seedling in  
Monterey, Calif. in 2000. The original seedling of the new  
cultivar was asexually propagated by stolons in a Nursery in  
Shasta County, Calif. Propagules were transplanted to a  
controlled breeding plot in Monterey county, Calif., where  
the variety was identified and selected for further evaluation.  
‘Driscoll Destin’ was subsequently asexually propagated  
and underwent further testing in Hillsborough county, Fla.  
for three years. This propagation and testing has demon-  
strated that the combination of traits disclosed herein which  
characterize the new variety are fixed and retained true to  
type through successive generations of asexual reproduc-  
tion.

**3. SUMMARY OF THE INVENTION**

The present invention relates to a new and distinct variety  
of strawberry named ‘Driscoll Destin.’ The variety is botani-  
cally identified as *Fragaria*×*ananassa*. The new variety is  
distinguished from other varieties by a number of charac-  
teristics as set forth in Tables 1–4.

**4. COMPARISON TO SIMILAR VARIETIES**

The varieties which we believe to be similar to ‘Driscoll  
Destin’ from those known to us are ‘Biscayne’ (U.S. Plant  
Pat. No. 12,186) and ‘Key Largo’ (U.S. Plant Pat. No.  
8,649). There are several characteristics of the new variety  
that are different from, or not possessed by ‘Biscayne’ and

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‘Key Largo’. The new variety has a shorter plant height,  
larger flower diameter, an orange-red external fruit color,  
greater yield, weak leaf glossiness, very dense stipule  
pubescence, and a small hollow fruit center.

5 ‘Driscoll Destin’ is distinguished from its maternal parent  
‘73D144’ by its larger average berry size and improved  
shipability. ‘Driscoll Destin’ is distinguished from its pater-  
nal parent ‘88E94’ by its larger average berry size, improved  
shipping ability, and partial everbearing in comparison to the  
10 full everbearing of ‘88E94’.

**5. BRIEF DESCRIPTION OF THE DRAWINGS**

15 The accompanying photographs show typical specimens  
of the new variety, including fruit, foliage and flowers, in  
color as nearly true as it is reasonably possible to make in  
color illustrations of these characteristics.

FIG. 1 shows the upper side of the leaves of the plant.

FIG. 2 shows the whole the plant.

20 FIG. 3 shows the under side and underside of the flowers.

FIG. 4 shows a close-up of the fruit.

FIG. 5 shows the fruit in longitudinal cross-section.

25 **6. DESCRIPTION OF THE NEW VARIETY**

The following detailed description of the new variety is  
based upon observations taken of plants and fruit grown in  
Hillsborough county, Fla., U.S.A. This description is in  
accordance with UPOV terminology. Observations of  
30 ‘Driscoll Destin’, ‘Biscayne’ and ‘Key Largo’ were taken in  
side by side comparison in the 2003–2004 winter season.  
Plants for observation were harvested from McArthur,  
Calif., and held in refrigerated storage until planting in  
Hillsborough county, Fla. in October 2003. Plants were  
grown in raised beds of soil under conditions typical of  
commercial strawberry production in central Florida. Fruits  
were harvested twice weekly for yield determination from  
November 2003 to March 2004. Measurements of plant,



flower, and fruit characteristics were made in January 2004, approximately four months after planting. Color designations, color descriptions, and other phenotypical descriptions may deviate from the stated values and descriptions depending upon variation in environmental, seasonal, climatic and cultural conditions. Colors are described and the most similar color designations are provided from The Royal Horticultural Society (R.H.S.) Colour Chart.

### 6.1 PROPAGATION

The new variety is principally propagated by way of stolons. Although propagation by stolons is presently preferred, other known methods of propagating strawberry plants may be employed.

### 6.2 CHARACTERISTICS OF THE NEW VARIETY

Information on the new variety is presented in Tables 1, 2, 3 and 4. In the tables, the flowers described are secondary flowers except where indicated. The fruit described is the secondary fruit on five month old plants. Fruit and flower measurements are an average of both primary and secondary fruit and flowers.

Table 1 provides information on the plant and fruit characteristics of the new variety 'Driscoll Destin' compared with characteristics of 'Biscayne' and 'Key Largo.' Table 2 provides additional information of the plant and fruit characteristics of the new variety 'Driscoll Destin' compared with characteristics of the varieties 'Biscayne' and 'Key Largo.' Table 3 provides reactions of the new variety to stresses, pests and diseases compared with reactions of the varieties 'Biscayne' and 'Key Largo.' Table 4 provides isozyme characteristics of the new variety as compared to that of the varieties 'Biscayne' and 'Key Largo.'

TABLE 1

| QUANTITATIVE COMPARISON OF 'DRISCOLL DESTIN',<br>'BISCAYNE,' AND 'KEY LARGO' |                                 |                             |                             |
|------------------------------------------------------------------------------|---------------------------------|-----------------------------|-----------------------------|
|                                                                              | 'Driscoll<br>Destin'            | 'Biscayne'                  | 'Key Largo'                 |
| <u>Plant Characteristics</u>                                                 |                                 |                             |                             |
| Height of Plant (cm)                                                         | 8.1                             | 10.0                        | 11.6                        |
| Spread of Plant (cm)                                                         | 32.4                            | 31.3                        | 32.1                        |
| Number of Crowns                                                             | 3.4                             | 3.2                         | 4.2                         |
| <u>Leaf Characteristics</u>                                                  |                                 |                             |                             |
| Terminal Leaflet<br>Length (cm)                                              | 8.2                             | 7.6                         | 8.4                         |
| Terminal Leaflet<br>Width (cm)                                               | 8.3                             | 8.1                         | 8.0                         |
| Terminal Leaflet<br>Length/Width Ratio                                       | 0.99                            | 0.93                        | 1.06                        |
| # Teeth/Terminal<br>Leaflet                                                  | 24.5                            | 27.0                        | 21.6                        |
| Color of upper side                                                          | 135A<br>medium to<br>dark green | 137A<br>light green         | 137A<br>light green         |
| Color of under side                                                          | 136C<br>light gray<br>green     | 139C<br>light gray<br>green | 139C<br>light gray<br>green |
| Petiole Length (cm)                                                          | 8.6                             | 9.2                         | 11.5                        |
| Petiole Color                                                                | 145B<br>yellow green            | 142D<br>yellow green        | 142D<br>yellow green        |
| Petiolule Length (mm)                                                        | 7.0                             | 5.5                         | 6.3                         |
| Petiolule Diameter<br>(mm)                                                   | 2.1                             | 2.2                         | 1.9                         |

TABLE 1-continued

| QUANTITATIVE COMPARISON OF 'DRISCOLL DESTIN',<br>'BISCAYNE,' AND 'KEY LARGO' |                                               |                                               |                                               |
|------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|
|                                                                              | 'Driscoll<br>Destin'                          | 'Biscayne'                                    | 'Key Largo'                                   |
| Petiolule Color                                                              | 145C<br>yellow green                          | 142D<br>yellow green                          | 142D<br>yellow green                          |
| Bract Frequency                                                              | 8%<br>typically<br>single                     | 83%<br>typically<br>paired                    | 0%                                            |
| Stipule Length (cm)                                                          | 3.8                                           | 3.7                                           | 3.4                                           |
| Stipule Width (cm)                                                           | 1.6                                           | 2.0                                           | 1.5                                           |
| <u>Stolon Characteristics</u>                                                |                                               |                                               |                                               |
| Anthocyanin color                                                            | 60A<br>red                                    | 59B<br>red                                    |                                               |
| Diameter at bract<br>(mm)                                                    | 3.59                                          | 3.24                                          | 3.30                                          |
| Avg. # of Daughter<br>plants (2003 Nursery)                                  | 51                                            | 72                                            | 52                                            |
| <u>Flower Characteristics</u>                                                |                                               |                                               |                                               |
| Petal Length (cm)                                                            | 1.5                                           | 1.36                                          | 1.65                                          |
| Petal Width (cm)                                                             | 1.3                                           | 1.49                                          | 1.46                                          |
| Petal Length/Width<br>Ratio                                                  | 1.09                                          | 0.91                                          | 1.13                                          |
| Flower Diameter (cm)                                                         | 3.0                                           | 2.76                                          | 2.72                                          |
| Calyx Diameter (cm)                                                          | 4.0                                           | 3.86                                          | 5.11                                          |
| Sepal Length (mm)                                                            | 15.8                                          | 15.3                                          | 21.5                                          |
| Sepal Width (cm)                                                             | 6.6                                           | 7.22                                          | 9.15                                          |
| Petal Color                                                                  | 155C white                                    | 155C white                                    | 155C white                                    |
| Receptical Color                                                             | 150A<br>yellow green                          | 150A<br>yellow green                          | 150A<br>yellow green                          |
| Anther Color                                                                 | 13A yellow                                    | 9A yellow                                     | 9A yellow                                     |
| Fruiting Truss Length<br>(cm)                                                | 8.8                                           | 8.7                                           | 13.6                                          |
| <u>Fruit Characteristics</u>                                                 |                                               |                                               |                                               |
| Fruit Length (cm)                                                            | 5.6                                           | 4.8                                           | 4.8                                           |
| Fruit Width (cm)                                                             | 4.3                                           | 4.0                                           | 3.5                                           |
| Fruit Length/Width<br>Ratio                                                  | 1.31                                          | 1.20                                          | 1.37                                          |
| Average Berry Weight<br>(g)                                                  | 28.9                                          | 26.5                                          | 23.2                                          |
| External Color                                                               | 45B<br>orange red                             | 46B<br>red                                    | 46B<br>red                                    |
| Internal Color                                                               | 179C<br>orange red                            | 44A<br>medium red                             | 44B<br>medium red                             |
| Achene Coloration                                                            | 184B to 150A<br>greyed red to<br>yellow green | 180B to 150C<br>greyed red to<br>yellow green | 182B to 150A<br>greyed red to<br>yellow green |
| Achenes per berry                                                            | 194                                           | 109                                           | 104                                           |
| Achene weight (g)                                                            | 0.000077                                      | 0.00046                                       | 0.00058                                       |
| 2003-2004 Fruit Yield<br>(g/plant)                                           | 285                                           | 243                                           | 251                                           |

TABLE 2

| QUALITATIVE COMPARISON OF 'DRISCOLL DESTIN',<br>'BISCAYNE,' AND 'KEY LARGO' |                      |              |                  |
|-----------------------------------------------------------------------------|----------------------|--------------|------------------|
|                                                                             | 'Driscoll<br>Destin' | 'Biscayne'   | 'Key Largo'      |
| <u>Plant</u>                                                                |                      |              |                  |
| Habit                                                                       | flat                 | flat globose | globose          |
| Canopy Density                                                              | dense                | medium       | medium           |
| Vigor                                                                       | strong               | strong       | medium           |
| <u>Leaf</u>                                                                 |                      |              |                  |
| Shape in cross section                                                      | concave              | concave      | slightly concave |
| Interveinal blistering                                                      | very strong          | weak         | weak             |
| Glossiness                                                                  | weak                 | medium       | medium           |
| Number of leaflets                                                          | three only           | three only   | three only       |



TABLE 2-continued

| QUALITATIVE COMPARISON OF ‘DRISCOLL DESTIN’,<br>‘BISCAYNE,’ AND ‘KEY LARGO’ |                                     |             |                  |
|-----------------------------------------------------------------------------|-------------------------------------|-------------|------------------|
|                                                                             | ‘Driscoll<br>Destin’                | ‘Biscayne’  | ‘Key Largo’      |
| Terminal leaflet<br>margin profile                                          | revolute to flat                    | revolute    | revolute         |
| Terminal leaflet shape<br>of base                                           | obtuse                              | rounded     | slightly oblique |
| Terminal leaflet shape<br>of teeth                                          | acute to obtuse                     | rounded     | rounded          |
| Stipule pubescence                                                          | very dense                          | medium      | medium           |
| Petiole pubescence                                                          | medium                              | medium      | medium           |
| Petiole pose of hairs                                                       | outwards                            | outwards    | upwards          |
| <u>Stolon</u>                                                               |                                     |             |                  |
| Anthocyanin<br>coloration                                                   | medium                              | strong      |                  |
| Thickness                                                                   | thick                               | medium      |                  |
| Pubescence                                                                  | medium                              | medium      |                  |
| <u>Inflorescence</u>                                                        |                                     |             |                  |
| Position relative to<br>foliage                                             | beneath                             | level       | level to above   |
| Diameter of calyx<br>relative to corolla on<br>secondary flowers            | same size                           | larger      | much larger      |
| Diameter of inner<br>calyx relative to outer<br>on secondary flowers        | same size to<br>larger              | same size   | smaller          |
| Spacing of petals                                                           | overlapping                         | overlapping | overlapping      |
| <u>Fruiting Truss</u>                                                       |                                     |             |                  |
| Attitude at first<br>picking                                                | prostrate                           | prostrate   | semi-erect       |
| <u>Fruit</u>                                                                |                                     |             |                  |
| Predominant shape                                                           | conical to<br>almost<br>cylindrical | conical     | conical          |
| Difference in shapes<br>between primary and<br>secondary fruits             | moderate                            | slight      | slight           |
| Band without achenes                                                        | medium                              | narrow      | narrow           |

TABLE 2-continued

| QUALITATIVE COMPARISON OF ‘DRISCOLL DESTIN’,<br>‘BISCAYNE,’ AND ‘KEY LARGO’ |                      |            |             |
|-----------------------------------------------------------------------------|----------------------|------------|-------------|
|                                                                             | ‘Driscoll<br>Destin’ | ‘Biscayne’ | ‘Key Largo’ |
| Unevenness of surface                                                       | weak                 | weak       | weak        |
| Evenness of color                                                           | uneven               | even       | even        |

6.4 ISOZYME ANALYSIS

In addition to the morphological description above, the new cultivar ‘Driscoll Destin’ has been analyzed to obtain an indication of its genetic makeup to provide further means for identifying the new variety and distinguishing it from other somewhat similar and/or related strawberry varieties. Specifically, leaf samples of ‘Driscoll Destin’, ‘Biscayne’ and ‘Key Largo’ were analyzed by electrophoresis for isozyme patterns of the enzymes phosphoglucosomerase (“PGI”), leucine aminopeptidase (“LAP”) and phosphoglucomutase (“PGM”). See J. Amer. Soc. Hort. Sci. 106:684–687. Isozyme characterization of the three varieties is presented in Table 4, with the letters representing the banding patterns for each enzyme as designated in the above-identified article.

TABLE 4

| ISOZYME ANALYSIS FOR ‘DRISCOLL DESTIN’,<br>‘BISCAYNE,’ AND ‘KEY LARGO’ |                      |            |             |
|------------------------------------------------------------------------|----------------------|------------|-------------|
| Locus                                                                  | ‘Driscoll<br>Destin’ | ‘Biscayne’ | ‘Key Largo’ |
| PGI                                                                    | A2                   | A1         | A1          |
| LAP                                                                    | B3                   | B3         | B3          |
| PGM                                                                    | C2                   | C4         | C4          |

What is claimed:

1. A new and distinct variety of strawberry plant, substantially as shown and described.

\* \* \* \* \*





**FIG.1**





FIG.2



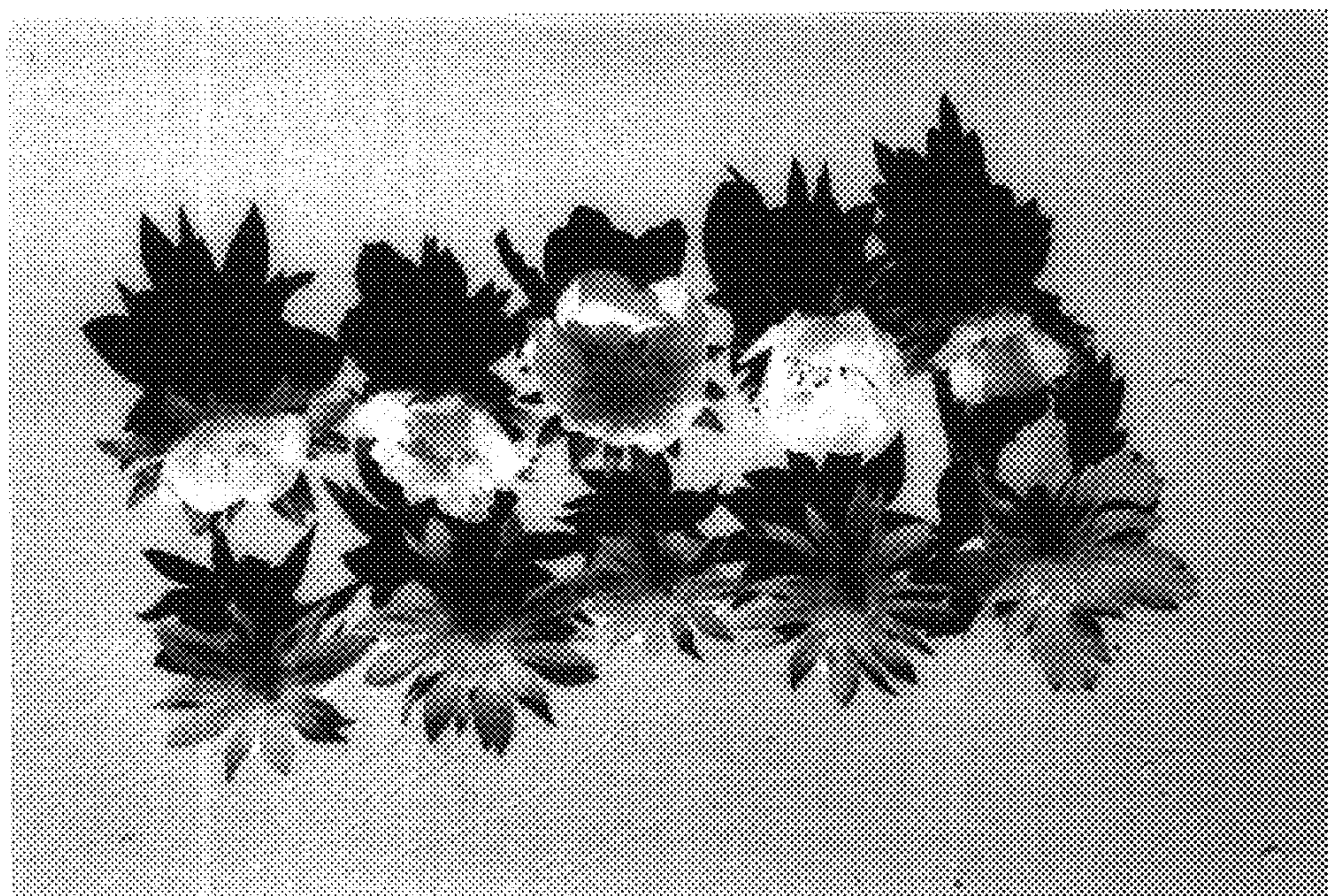


FIG.3



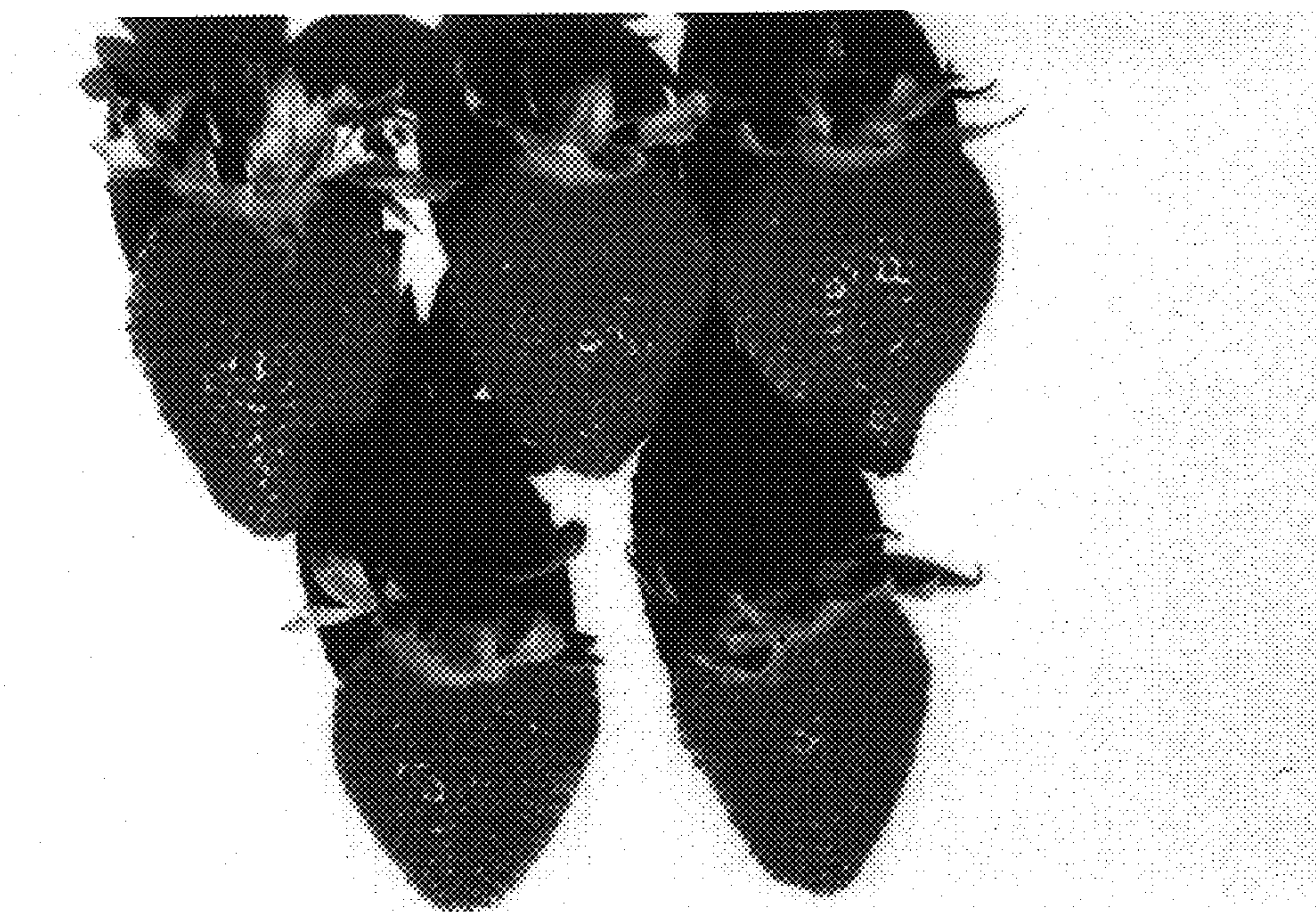


FIG.4





FIG.5